

PRIYADHARSHAN SENGUTUVAN

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Master of Science in Data Science

May 2026

Courses: Machine Learning, NLP, Algorithms, MLOps

Sri Krishna College of Engineering and Technology

Coimbatore, India

Bachelor of Technology in Information Technology

May 2023

SKILLS

Programming: Python (NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow), SQL, Java, R, C#

Data Engineering: ETL Pipelines, Apache Airflow, GCP BigQuery, Data Modeling, Data Quality, Docker, Git

Analytics: A/B Testing, Statistical Analysis, Data Visualization, Tableau, Business Intelligence, Product Analytics

Machine Learning: Supervised/Unsupervised Learning, XGBoost, Feature Engineering, Model Evaluation, MLOps

Frameworks: FastAPI, LangChain, Prometheus, Grafana, MLflow, CI/CD

PROFESSIONAL EXPERIENCE

AriesView

Boston, MA (Remote)

AI/ML Engineer Intern

Sept 2025 - Dec 2025

- Led full-stack development of analytics dashboards processing 500+ properties daily, enabling 1K+ investors to evaluate NOI, cap rate, and DSCR metrics, accelerating investment decisions by 60%.
- Improved data accuracy from 85% to 95% by implementing OCR validation pipelines with automated quality checks, eliminating manual entry errors across financial records processing.
- Increased RAG chatbot query accuracy from 68% to 87% through systematic A/B testing of embedding strategies and search architectures, reducing average support resolution time by 40%.

Psiog Digital Private Limited

Chennai, India

Software Engineer Intern

Feb 2023 - Dec 2023

- Partnered with business stakeholders to design and deploy React/C# dashboards displaying critical KPIs, replacing manual Excel workflows and enabling data-driven decisions across finance and operations teams.
- Performed data analysis on transaction patterns and optimized SQL Server through query rewriting and indexing, reducing analytics query load times by 75% and improving data accessibility for 50+ business users.
- Designed and implemented REST APIs connecting previously siloed backend services, streamlining data access and improving cross-functional team productivity by consolidating data sources.

PROJECTS

RAG-Based Operations Assistant for E-Commerce ([GitHub](#))

Python, LangChain, GCP, Airflow, Docker

- Architected an end-to-end data pipeline ingesting seller support data into BigQuery using Airflow ETL processes with incremental loading, enabling real-time analytics dashboards for operations team.
- Built RAG chatbot using LangChain and GPT-4 to automate seller support queries, improving response accuracy from 68% to 87% through systematic experimentation with embeddings, chunking strategies, and prompt engineering.
- Established a data quality framework ensuring semantic search reliability and maintaining data freshness for 100K+ product records.

Customer Churn Analytics & Prediction System ([GitHub](#))

Python, FastAPI, Docker, Prometheus

- Performed exploratory data analysis on 7K+ telecom records identifying three high-risk customer segments representing 35% of churn, enabling targeted retention strategies and informing product improvements.
- Deployed production ML system with A/B testing framework comparing Random Forest and Gradient Boosting models, optimizing latency by 84% (36ms to 5.7ms) while maintaining comparable accuracy (0.84 vs 0.83 AUC).
- Implemented monitoring infrastructure with Prometheus and Grafana tracking data pipeline health, prediction volume, and error rates across containerized services to ensure data quality and system reliability.

Financial Research Assistant ([GitHub](#))

Python, Llama 3.1, ChromaDB, Ollama

- Built a local LLM-powered financial research tool analyzing SEC 10-K filings using Llama 3.1 via Ollama and ChromaDB vector storage, enabling instant querying of 100+ page documents with cited sources, eliminating hours of manual analysis for investment research.
- Achieved 60% performance improvement by optimizing chunking strategies (2000-char vs 1000-char) and deploying local inference architecture, reducing query latency from 3+ minutes to 34 seconds while maintaining accuracy.
- Engineered end-to-end data pipeline integrating SEC Edgar API for automated 10-K ingestion, Sentence Transformers for semantic embeddings, and citation tracking—demonstrating production-ready financial intelligence tooling applicable to due diligence and compliance monitoring.